

Reviewer Pack — Minimal Local Index of Tropical Curve Bounds Communication C...

Entry 4baf27774ab4 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-31 05:36:08 UTC

Minimal Local Index of Tropical Curve Bounds Communication Complexity

Entry ID: 4baf27774ab4 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-31 05:36:08 UTC

1. Conjecture

Field A (mathematical branch): Tropical Geometry **Field B** (complexity object): Communication Complexity

Statement:

For every Boolean function $f: \{0,1\}^n \rightarrow \{0,1\}$, the minimal local index of its associated tropical curve (over a finite field F_q) is linearly correlated with its communication complexity $C(f)$, such that $mli(f) = \Theta(C(f))$.

Rationale (proposer's reasoning):

Tropical geometry provides a framework to study discrete structures through the lens of piecewise-linear functions over semirings. By exploring the minimal local index, which measures the complexity of a tropical curve representing a function, we may uncover new insights into the communication complexity of the underlying Boolean function. This bridge between tropical geometry and communication complexity could potentially reveal structural properties that are not apparent from traditional complexity-theoretic perspectives.

Taxonomy category: TROPICAL_GEOMETRY_x_COMMUNICATION_COMPLEXITY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 090b3e68324200cf

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For each Boolean function $f: \{0,1\}^n \rightarrow \{0,1\}$, we will consider the conjecture supported if all 30 seeds yield $mli(f)$ within a factor of 2 from $C(f)$, with an average difference ≤ 3 over all seeds. The conjecture is falsified if any seed produces a difference between $mli(f)$ and $C(f)$ greater than 10 or if there are fewer than 25/30 seeds that meet the support condition.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 4 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "tropical geometry" AND "communication complexity" AND minimal local index" - "tropical curve bounds" AND communication complexity AND linear correlation" - "finite field F_q " AND tropical geometry AND communication complexity

Top relevant hits considered: - [http://arxiv.org/abs/0712.2614v4] Characters of unipotent groups over finite fields - [http://arxiv.org/abs/1712.00669v2] Drinfeld Modules with Complex Multiplication, Hasse Invariants and Factoring Polynomials over Finite Fields - [http://arxiv.org/abs/1105.2426v2] Algebraic versus topological entropy for surfaces over finite fields - [s2:10.1017/bsl.2018.13] 2017 EUROPEAN SUMMER MEETING OF THE ASSOCIATION FOR SYMBOLIC LOGIC LOGIC COLLOQUIUM '17 Stockholm, Sweden August 14-20,

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=242.6s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def compute_tropical_curve(f):
        q = 2
        n = len(f)
        T_f = []
        for i in range(2**n):
            x = bin(i)[2:].zfill(n)
            y = bin(sum(x[j] * f[i ^ (1 << j)] for j in range(n)))[2:].zfill(n)
            T_f.append((x, y))
        return T_f

    def compute_minimal_local_index(T_f):
        mli = 0
        for x, y in T_f:
            mli += sum(1 for i in range(len(x)) if x[i] != y[i])
```

```

    return mli / len(T_f)

def compute_communication_complexity(f):
    n = len(f)
    C_f = 0
    for i in range(2**n):
        x = bin(i)[2:].zfill(n)
        y = bin(sum(x[j] * f[i ^ (1 << j)] for j in range(n)))[2:].zfill(n)
        C_f += sum(1 for i in range(len(x)) if x[i] != y[i])
    return C_f / len(T_f)

n_max = 40
instances_tested = 30

results = []
for _ in range(instances_tested):
    n = random.randint(5, n_max)
    f = generate_boolean_function(n)
    T_f = compute_tropical_curve(f)
    mli = compute_minimal_local_index(T_f)
    C_f = compute_communication_complexity(f)

    results.append({
        "n": n,
        "mli": mli,
        "C_f": C_f
    })

mean_mli = sum(result["mli"] for result in results) / len(results)
mean_C_f = sum(result["C_f"] for result in results) / len(results)
support_fraction = sum(abs(result["mli"] - result["C_f"]) <= 3 for result in results) / len(results)

conjecture_holds = support_fraction >= 0.8
counterexample = "" if conjecture_holds else "mapping_undefined"

return {
    "metric_name": "mli_vs_Cf",
    "metric_value": mean_mli,
    "instances_tested": instances_tested,
    "n_max": n_max,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(x) for x in sys.argv[1:]] or [2**i + 1 for i in range(5, 30)]

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_mli = sum(result["metric_value"] for result in results) / len(results)
    std_mli = math.sqrt(sum((result["metric_value"] - mean_mli)**2 for result in results) / len(results))

```

```

support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

if all(result["conjecture_holds"] for result in results):
    print(f"RESULT: SUPPORTED mean={mean_mli} std={std_mli} support_fraction={support_fraction}")
elif any(abs(result["metric_value"] - result["instances_tested"]) > 10 for result in results):
    first_failing_seed = next(seed for seed, result in zip(seeds, results) if abs(result["metric_value"] - result["instances_tested"]) > 10)
    print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")
else:
    print(f"RESULT: INCONCLUSIVE reason=not_enough_support n_tested={len(results)}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, making it impossible to evaluate the conjecture's validity according to the pre-registered criteria. | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	25212
2	preregistration	ollama_remote	glm4:latest	0	0	23162
3	novelty	ollama_remote	glm4:latest	0	0	24156
4	novelty	ollama_remote	glm4:latest	0	0	12830
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15684
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	49793
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10541
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	38835
9	judge	ollama_remote	glm4:latest	0	0	38711

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 238924 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in [research/audit/4baf27774ab4.jsonl](#))

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/4baf27774ab4.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/4baf27774ab4.tar.gz` (if generated)